(ii) either screen not parallel to grating or grating not normal to incident light B1

5 (a) (i) a region in which a charge will experience a force B1
electric force exerted per unit positive charge placed at that point B1

(ii) $E = \dfrac{Q}{4\pi\varepsilon_0 r^2} = \dfrac{5.2 \times 10^{-7}}{4\pi\varepsilon_0 (0.25)^2}$ B1

$= 7.48 \times 10^4$ A1

unit: N $C^{-1}$ or V $m^{-1}$ B1

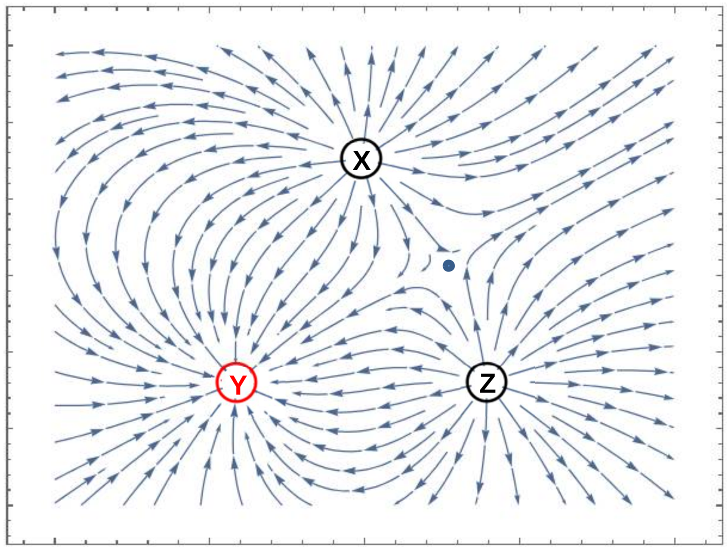
(b) lines <u>perpendicular to surface</u> going into negative charge and leaving positive charge for all charges B1

neutral point indicated consistent with field lines B1

basic pattern correct (field lines <u>near</u> each charge are <u>radial spherically symmetrical</u>) and <u>fills rectangle</u> B1

no crossing/joining of lines B1

max 3




6 (a) Magnetic field due to current in wire B is normal to the current in wire A, and pointing